Module 3 slides: Intravenous Infusions (worked examples)

Module 3 - Intravenous Infusions · Lecture slides, worked examples · 2 Sep 2026

Name _____ Date _____

Example 1 Theophylline is given by continuous intravenous infusion at 50 mg/hr. Its half-life is 6 hours and its volume of distribution is 30 L. What steady-state plasma concentration is expected?

Example 2 The same drug has a half-life of 6 hours and a volume of distribution of 30 L. What infusion rate would achieve a steady-state concentration of 20 mg/L?

A patient received an intravenous infusion of 150 mg over a period of 6 hours. The drug has an elimination rate constant of 0.231 hr^{-1} and an apparent volume of

distribution of 15 L.

Example 6 What is the plasma concentration at the end of the infusion?

Example 6 What is the concentration of the drug in the body 3 hours after cessation of the infusion?